

BPHO ROUND 0 · MARKSCHEME

Drill bank section 5

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

Original questions written for this course, in the Round 0 format. Not BPhO material.

Quick key

Q1 A	Q2 C	Q3 E	Q4 B	Q5 D
Q6 D	Q7 A	Q8 C	Q9 E	Q10 B
Q11 B	Q12 D	Q13 A	Q14 C	Q15 E
Q16 E	Q17 B	Q18 D	Q19 A	Q20 C
Q21 C	Q22 E	Q23 B	Q24 D	Q25 A

Worked solutions

Q1 · Dimensional analysis: which combination has the dimensions of pressure — **Answer A**

What is being tested. Whether you can reduce a compound quantity to its base dimensions and recognise that pressure is force per area.

Step 1 — the dimensions of pressure. Force is mass times acceleration, so $[F] = MLT^{-2}$, and area is L^2 . Therefore

$$[p] = [F] / [A] = MLT^{-2} / L^2 = ML^{-1} T^{-2}$$

Step 2 — test each option. Energy is $ML^2 T^{-2}$, so energy over volume $(ML^2 T^{-2})/L^3 = ML^{-1} T^{-2}$ -- exactly the dimensions of pressure. Power is $ML^2 T^{-3}$, so power over area is MT^{-3} ; force over length is MT^{-2} ; momentum over volume is $ML^{-2} T^{-1}$; and mass over area-time is $ML^{-2} T^{-1}$. Only the first matches. So **Answer: A**.

The distractors.

- **power / area** gives MT^{-3} : it carries a time to the minus three instead of minus two, the difference between a flux of energy and an actual pressure.
- **force / length** gives MT^{-2} , a force per unit length such as surface tension, short by one length.
- **momentum / volume** gives $ML^{-2} T^{-1}$, a momentum flux, again missing a length.
- **mass / (area x time)** is the same wrong dimensions as the momentum trap, a coincidence that separates the careful from the careless.

Relevant topics: base dimensions; reducing compound quantities; pressure as force per unit area.

The trap. Treating pressure as a force divided by a length (surface-tension dimensions) instead of by an area. The extra length is the difference between the right answer and three of the five options.

Q2 · A hinged rod held by a cable: the tension from moments — **Answer C**

What is being tested. Whether you can take moments about a hinge to dispose of the unknown reaction, and whether you resolve the cable correctly -- only its vertical part turns the rod about **A**.

Step 1 — the cable's direction. The rod is 4.0 m out, the cable rises 3.0 m to the wall, so the cable, the rod and the wall make a 3-4-5 triangle. The tension therefore has a vertical component $T_y = T \times 3/5$ and a horizontal component $T \times 4/5$.

Step 2 — choose the pivot. About the hinge **A** the two hinge reactions have zero moment arm, so they vanish from the equation. The only unknown left is **T**.

Step 3 — the clockwise moments. The 150 N load sits at the midpoint, 2.0 m from **A** : moment $150 \times 2.0 = 300 \text{ N m}$. The 45 N load hangs at **B**, 4.0 m from **A** : moment $45 \times 4.0 = 180 \text{ N m}$. Total clockwise = 480 N m .

Step 4 — the anticlockwise moment. Only the vertical component of the cable turns the rod, and it acts at **B** (4.0 m from **A**):

$$T_y \times 4.0 = (3T/5) \times 4.0 = 12T/5$$

Step 5 — solve.

$$12T/5 = 480$$

$$T = 480 \times 5 / 12 = 200 \text{ N}$$

So **Answer: C**.

Sanity check. The cable pulls at an angle, so its full tension must exceed the 150 N and 45 N loads that it balances between them; 200 N is just above that, which is the right direction. The 5.0 m cable length never appears, because only the 3.0 m rise does any vertical work.

The distractors.

- **150 N** is the mid-rod load on its own -- a candidate who balances only that and forgets the end load and the cable angle lands here.
- **250 N** uses the full 5.0 m cable as the tension's moment arm, treating the cable as if it pulled straight down.
- **120 N** divides the total moment by the 4.0 m rod but omits the 3/5 vertical factor, so it under-values the cable.
- **160 N** divides by the horizontal arm but forgets to multiply the vertical component by the 4.0 m distance from the hinge.

Relevant topics: moments about a pivot; resolving a tension; the 3-4-5 triangle; why the hypotenuse is a distraction.

The trap. Using the 5.0 m cable length as the moment arm of the tension. Only the 3.0 m vertical rise (the 3/5 component) does any turning about the hinge; the hypotenuse never enters the equation.

Q3 · Two potential dividers: the voltmeter reading between their midpoints — **Answer E**

What is being tested. Whether you treat each branch as an independent potential divider and then take the difference of two internal potentials -- not a drop across a single resistor.

Step 1 — the left branch current. Its total resistance is $2.0 + 6.0 = 8.0$ ohm, so

$$I_{\text{left}} = 12 / 8.0 = 1.5 \text{ A}$$

Step 2 — the potential of X. X is the midpoint; below it sits the 6.0 ohm resistor, across which the drop is $1.5 \times 6.0 = 9.0$ V. So X is 9.0 V above the bottom rail, equivalently $12 - 9.0 = 3.0$ V below the top -- but the relevant number is $V_X = 9.0$ V above the rail.

Step 3 — the right branch current. Its total resistance is $6.0 + 3.0 = 9.0$ ohm :

$$I_{\text{right}} = 12 / 9.0 = 4/3 \text{ A}$$

Step 4 — the potential of Y. Below Y sits the 3.0 ohm resistor, drop $(4/3) \times 3.0 = 4.0$ V. So $V_Y = 4.0$ V above the bottom rail.

Step 5 — the meter reading. An ideal voltmeter draws no current, so it simply reads the potential difference between X and Y:

$$V = V_X - V_Y = 9.0 - 4.0 = 5.0 \text{ V}$$

So **Answer: E.**

The distractors.

- **3.0 V** is the drop across the 2.0 ohm resistor on the left branch -- the distance from X up to the top rail, not the meter reading.
- **8.0 V** is the drop across the 6.0 ohm resistor on the right branch.
- **9.0 V** is the potential of X above the rail; a candidate who forgets to subtract Y reads the supply-split instead of the difference.
- **1.5 V** is merely the left branch current, written as though it were a voltage.

Relevant topics: potential dividers; branch currents; an ideal voltmeter reads a potential difference.

The trap. Reading the drop across a single resistor (the 2.0 ohm or the 6.0 ohm) instead of the potential difference X minus Y. The meter bridges two internal points, so both midpoints must be found first and then subtracted.

Q4 · Order of magnitude: the number of seconds in one year — **Answer B**

What is being tested. A clean order-of-magnitude estimate: round each factor sensibly and read the power of ten, without a calculator.

Step 1 — express a year in seconds.

$$\begin{aligned}
 1 \text{ year} &= 365 \text{ days} \times 24 \text{ hours} \times 3600 \text{ seconds} \\
 &= 31536000 \text{ s}
 \end{aligned}$$

Step 2 — round to one significant figure. The product is about 3.2×10^7 seconds. The nearest power of ten to 3.2×10^7 is 10^7 (it would need to pass $3.2 \times 10^7 \rightarrow 10^{7.5}$ to round the other way). So **Answer: B**.

The distractors.

- 10^5 is the order of the number of seconds in a day (86400), not a year.
- 10^6 is the number of seconds in an hour, a thousand times too small.
- 10^8 is about a hundred times too large, as if a year were closer to a century.
- 10^9 is a thousand times too large, as if the year were a millennium.

Relevant topics: order-of-magnitude estimation; powers of ten; the pi-times-ten-to-the-seven seconds-per-year rule.

The trap. Confusing the scales of a day (10^5 s), an hour (10^6 s) and a year (10^7 s). The day and the hour are the two that tempt a candidate who multiplies wrongly.

Q5 · A velocity-time graph: the total distance from the area — **Answer D**

What is being tested. The single most-used fact about a velocity-time graph: the area beneath it is the distance, read off the drawn axes.

Step 1 — identify the three regions. From the graph: velocity rises linearly from 0 to 20 m/s over the first 5 s; stays at 20 m/s until 15 s; falls linearly to 0 by 19 s. So the area is a triangle, then a rectangle, then a triangle.

Step 2 — the first triangle (0 to 5 s). Its area is $A1 = \frac{1}{2} \times 5 \times 20 = 50 \text{ m}$.

Step 3 — the rectangle (5 to 15 s). The cruise lasts $15 - 5 = 10 \text{ s}$ at 20 m/s:

$$A2 = 10 \times 20 = 200 \text{ m}$$

Step 4 — the second triangle (15 to 19 s). The braking phase lasts $19 - 15 = 4 \text{ s}$:

$$A3 = \frac{1}{2} \times 4 \times 20 = 40 \text{ m}$$

Step 5 — add them.

$$\text{distance} = 50 + 200 + 40 = 290 \text{ m}$$

So **Answer: D**.

The distractors.

- **250 m** double-counts the braking triangle or tacks it onto the rectangle twice.

- **300 m** treats the whole 15 s as a 20 m/s rectangle (15×20) and ignores that the first and last phases are triangles.
- **200 m** is the cruise rectangle only.
- **340 m** invents an extra triangle that is not on the graph.

Relevant topics: v-t graphs; area = distance; splitting into triangles and rectangles.

The trap. Taking the cruise rectangle (200 m) as the whole answer and ignoring the two triangular end regions. Each triangle is real distance, and only their sum is the journey.

Q6 · A block on a table linked over a pulley: the normal reaction — **Answer D**

What is being tested. Whether you resolve forces on the correct body. The normal reaction on the table block balances only the vertical forces on that block, and the string is horizontal, so the hanging mass is irrelevant to it.

Step 1 — the forces on the 5.0 kg block. Vertically: its weight mg acts down, the normal reaction N acts up. Horizontally: the string tension acts sideways. There is no vertical acceleration, so the vertical forces balance on their own.

Step 2 — the string is horizontal. The pulley turns the string through a right angle; at the table block the string is along the table, so its tension has no vertical component. The hanging 3.0 kg block therefore contributes nothing to N .

Step 3 — balance vertically.

$$N - mg = 0$$

$$N = 5.0 \times 9.8 = 49 \text{ N}$$

So **Answer: D.**

Step 4 — check. The reaction equals the block's own weight. A common error is to add the hanging mass or to substitute its weight; neither belongs here because the hanging block's pull is horizontal.

The distractors.

- **20 N** comes from a wrong value of g or from taking only part of the weight.
- **78 N** adds both masses, $(5.0 + 3.0) \times 9.8$, as if the hanging block also rested on the table.
- **29 N** is the hanging block's weight, 3.0×9.8 , wrongly applied to the table block.
- **50 N** uses $g = 10$ on the 5.0 kg mass and rounds up.

Relevant topics: resolving forces; normal reaction; which body a force acts on.

The trap. Adding the hanging 3.0 kg block to the 5.0 kg block, or using the hanging weight, when finding the normal reaction. The string is horizontal, so the hanging mass's weight does not touch the table's normal force.

Q7 · A vertical circle: the minimum speed at the bottom for the string to stay taut — **Answer A**

What is being tested. The classic vertical-circle condition: the string stays taut only if, at the top, gravity alone can supply the centripetal force. Energy then connects that top speed to the launch speed at the bottom.

Step 1 — the taut-string condition at the top. At the highest point the tension is smallest. The string is just taut when $T = 0$, so weight provides all the centripetal force:

$$m v_{\text{top}}^2 / r = m g \Rightarrow v_{\text{top}}^2 = g r$$

Step 2 — energy from bottom to top. The top is $2r$ higher than the bottom, so with no work lost:

$$\frac{1}{2} m u^2 = \frac{1}{2} m v_{\text{top}}^2 + m g (2r)$$

Step 3 — substitute and solve for u. Divide by m and put in $v_{\text{top}}^2 = g r$:

$$\begin{aligned} \frac{1}{2} u^2 &= \frac{1}{2} g r + 2 g r = \left(\frac{1}{2} + 2\right) g r = \left(\frac{5}{2}\right) g r \\ u^2 &= 5 g r \\ u &= \sqrt{5 g r} \end{aligned}$$

So **Answer: A.**

Step 4 — check the size. The answer must exceed both $\sqrt{g r}$ (the top speed) and $\sqrt{4 g r}$ (the speed from a $2r$ drop alone), because the particle needs enough speed to climb the two radii AND keep the string taut. $\sqrt{5 g r}$ does that.

The distractors.

- $\sqrt{2 g r}$ uses only the energy gain $2 g r$ as the speed, forgetting that the top still needs $g r$ of centripetal speed.
- $\sqrt{3 g r}$ counts three radii of rise instead of two.
- $\sqrt{g r}$ is the top speed with the particle momentarily at rest there -- which is exactly when the string goes slack.
- $\sqrt{4 g r}$ is the speed gained in falling $2r$, ignoring the centripetal requirement at the top.

Relevant topics: vertical circular motion; the taut-string condition; conservation of mechanical energy.

The trap. Forgetting that at the top the particle still needs $v_{\text{top}}^2 = g r$ for the string to stay taut, and giving $\sqrt{2gr}$ or $\sqrt{4gr}$ from energy alone. The $5gr$ combines a $2r$ climb with the top's own centripetal need.

Q8 · A wave snapshot: the transverse velocity from the local slope — **Answer C**

What is being tested. The link between a snapshot (which shows slope) and motion (which is velocity): for a right-travelling wave the transverse velocity is minus the wave speed times the local slope.

Step 1 — the velocity-slope relation. A wave moving right has the form $y = f(x - v t)$. Differentiating with respect to time at fixed x :

$$dy/dt = -v (dy/dx)$$

so the transverse velocity is the wave speed times the slope, with a minus sign.

Step 2 — read the slope at Q. The stem gives the slope at Q as -0.40 . In the snapshot Q lies on the descending flank between the crest and the trough, so the string falls as x increases -- a negative slope, as stated.

Step 3 — compute.

$$v_y = -1.5 \times (-0.40) = +0.60 \text{ m/s}$$

A positive transverse velocity is upward. So **Answer: C**.

Step 4 — check the direction. On a right-travelling wave a point on the descending side is being lifted toward the crest that is approaching from the left, so it moves up -- consistent with the positive sign.

The distractors.

- **0.60 m/s downward** keeps the magnitude but drops the minus sign of the relation, sending the point the wrong way.
- **0.40 m/s upward** multiplies the speed by the slope with no minus, then loses the factor of 1.5.
- **0.90 m/s upward** adds the speed and the slope as if they were a sum.
- **0 m/s** is the velocity at a crest or a trough, where the slope is zero, not at Q.

Relevant topics: transverse waves; $v_y = -v (dy/dx)$; reading slope from a snapshot.

The trap. Dropping the minus sign in $v_y = -v(dy/dx)$, so a downward-sloping point (Q, on the descending flank) is sent downward instead of upward. The sign is the entire question.

Q9 · A converging lens: an object inside the focal length gives a virtual image — **Answer E**

What is being tested. The lens equation when the object is inside the focal length -- the case that yields a virtual, magnified image -- and the sign convention that makes v come out negative.

Step 1 — the lens equation. With the usual convention (real object and image distances positive, converging lens f positive):

$$1/u + 1/v = 1/f$$

Step 2 — substitute. Here $u = 12 \text{ cm}$ and $f = 20 \text{ cm}$. Note $u < f$, so the object is inside the focal length.

$$1/v = 1/f - 1/u = 1/20 - 1/12$$

Step 3 — the arithmetic.

$$1/20 - 1/12 = 3/60 - 5/60 = -2/60 = -1/30$$

so $v = -30 \text{ cm}$

The negative sign means the image is virtual and on the same side of the lens as the object.

Step 4 — the magnification.

$$m = -v/u = -(-30)/12 = 30/12 = 2.5$$

So **Answer: E.**

Step 5 — check. A virtual image is upright (m positive) and, for an object inside the focus, enlarged ($m > 1$). 2.5 satisfies both, which is the right physical sense.

The distractors.

- **0.40** is u/v in magnitude, the reciprocal of the magnification -- the answer turned upside down.
- **1.7** is $20/12$, the ratio of focal length to object distance, which is not the magnification.
- **2.0** comes from a mis-rearranged equation such as $f/(f - u)$.
- **0.60** is $12/20$, the ratio of object distance to focal length.

Relevant topics: the lens equation; the virtual image for $u < f$; magnification $m = -v/u$.

The trap. Swapping u and v to get 0.40, or using f/u and u/f , when the magnification is $-v/u$ with a virtual (negative) v . The negative v is the signal that the image is enlarged, not diminished.

Q10 · Two opposing cells: the current in the single loop — **Answer B**

What is being tested. Two cells facing opposite ways subtract, and internal resistances add to the external resistor. The loop is then one net emf over one total resistance.

Step 1 — the net emf. The 12 V cell has its positive terminal up and the 6 V cell has its positive terminal down, so around the loop they drive in opposite senses. The 12 V cell is larger, so:

$$E_{\text{net}} = 12 - 6 = 6 \text{ V}$$

Step 2 — the total resistance. The resistor and both internal resistances are in series:

$$R_{\text{total}} = 3.0 + 1.0 + 2.0 = 6.0 \text{ ohm}$$

Step 3 — the current.

$$I = E_{\text{net}} / R_{\text{total}} = 6 / 6.0 = 1.0 \text{ A}$$

Step 4 — the voltage shares. The 6.0 ohm total means each ohm carries the 1.0 A, so the two internal resistances ($1.0 + 2.0 = 3.0$ ohm) drop $3.0 \times 1.0 = 3.0 \text{ V}$, leaving $6.0 - 3.0 = 3.0 \text{ V}$ across the external 3.0 ohm resistor -- consistent with $3.0 \times 1.0 = 3.0 \text{ V}$.

$$V_{\text{internal}} = (1.0 + 2.0) \times 1.0 = 3.0 \text{ V} ; V_{\text{load}} = 3.0 \times 1.0 = 3.0 \text{ V}$$

So **Answer: B.**

Step 5 — check the sense. The net emf is positive in the direction of the 12 V cell, so the current circulates that way, and its modest size (1.0 A) reflects that the two cells are largely cancelling each other.

The distractors.

- **2.0 A** drives both emfs the same way: $(12 + 6)/9 = 2.0$ -- adding when they oppose.
- **3.0 A** adds the emfs and uses all three resistances but with the wrong combination $18/6$.

- **0.50 A** divides the net 6 V by only the external 3.0 ohm, forgetting the internal resistances.
- **6.0 A** mistakes the net emf of 6 V for the current in amperes.

Relevant topics: cells in series opposing; internal resistance; Kirchhoff's loop law.

The trap. Adding the two emfs because both are batteries, instead of subtracting them, or leaving out the internal resistances. The net drive is $12 - 6 = 6$ V over 6.0 ohm total.

Q11 · Radioactive decay: the fraction remaining after a number of half-lives — **Answer B**

What is being tested. The geometric nature of radioactive decay: after each half-life the remaining amount is halved, so after n half-lives it is $(1/2)^n$.

Step 1 — how many half-lives?

$$n = 24 \text{ days} / 8.0 \text{ days} = 3$$

Step 2 — apply the halving three times.

$$\text{fraction} = (1/2)^3 = 1/8$$

Step 3 — follow the decay one step at a time. Rather than jump to the cube, watch each halving: after the first half-life the fraction is $1/2$, after the second it is $(1/2) \times (1/2) = 1/4$, after the third it is $(1/4) \times (1/2) = 1/8$.

$$1/2 \rightarrow 1/4 \rightarrow 1/8 \text{ over three halvings}$$

Step 4 — confirm the time divides exactly. 24 days is three exact 8.0-day intervals with no remainder, so there is no partial fourth half-life to account for; the fraction is the clean value $(1/2)^3$.

So **Answer: B**.

Step 5 — check the bracket. Three halvings must give something below $1/2$ (one halving) but above $1/16$ (four halvings). $1/8$ sits exactly there.

The distractors.

- **$1/4$** is the result of only two half-lives, as if $24/8$ had been read as two.
- **$1/16$** is four half-lives -- one too many.
- **$3/8$** subtracts the three half-lives from 1, treating decay as a linear loss.
- **$1/2$** is the fraction after a single half-life, the start of the process rather than its end.

Relevant topics: half-life; geometric decay; the $(1/2)^n$ rule.

The trap. Treating the three half-lives as a subtraction from 1 (giving $3/8$) instead of three multiplications by $1/2$. Decay is geometric, so each half-life multiplies the remainder by $1/2$.

Q12 · The momentum of a photon of given wavelength — **Answer D**

What is being tested. The photon momentum formula $p = h/\lambda$, and keeping it distinct from the photon energy $E = hc/\lambda$.

Step 1 — the formula. A photon of wavelength λ has momentum

$$p = h / \lambda$$

Step 2 — the wavelength in metres.

$$600 \text{ nm} = 600 \times 10^{-9} \text{ m} = 6.0 \times 10^{-7} \text{ m}$$

Step 3 — substitute.

$$p = 6.6 \times 10^{-34} / (6.0 \times 10^{-7}) = 1.1 \times 10^{-27} \text{ kg m s}^{-1}$$

So **Answer: D.**

Step 4 — dimensional check. $(\text{J s})/\text{m} = (\text{kg m}^2 \text{ s}^{-2} \text{ s})/\text{m} = \text{kg m/s}$, which is a momentum. That confirms the formula, not the energy.

The distractors.

- 3.3×10^{-19} is the photon energy hc/λ , the wrong quantity.
- 5.5×10^{-28} loses a factor of ten in the nanometre-to-metre conversion.
- 1.1×10^{-19} uses the energy hf without dividing by c , massively too large.
- 6.6×10^{-34} is just Planck's constant with the wavelength ignored.

Relevant topics: photon momentum; $p = h/\lambda$; the difference between photon energy and momentum.

The trap. Computing the photon energy hc/λ and offering it as the momentum, or forgetting to convert 600 nm to 6.0×10^{-7} m. Momentum is h/λ , not hc/λ .

Q13 · Converting a speed from m/s to km/h — **Answer A**

What is being tested. A two-step unit conversion: get the speed in m/s, then change to km/h with the factor 3.6.

Step 1 — the speed in m/s.

$$v = 100 \text{ m} / 10 \text{ s} = 10 \text{ m/s}$$

Step 2 — convert to km/h. One metre per second is $(1/1000 \text{ km})/(1/3600 \text{ h}) = 3.6 \text{ km/h}$. So:

$$v = 10 \times 3.6 = 36 \text{ km/h}$$

So **Answer: A.**

The distractors.

- **10 km/h** is the speed in m/s left unconverted.
- **3.6 km/h** divides by 3.6 instead of multiplying.

- **360 km/h** multiplies the 100 and 10 together as if they made the speed.
- **100 km/h** offers the distance as the speed.

Relevant topics: average speed; the m/s to km/h conversion factor 3.6.

The trap. Leaving the answer as 10 m/s, or dividing by 3.6 instead of multiplying. From m/s to km/h you multiply by 3.6 (1000 m per km, 3600 s per hour).

Q14 · A horizontal projectile: the range from the time of fall — **Answer C**

What is being tested. The independence of horizontal and vertical motion for a projectile launched horizontally: the fall time comes from the height, the range from the horizontal speed.

Step 1 — the time of fall. Vertically the ball starts with no speed and drops 20 m under gravity:

$$h = \frac{1}{2} g t^2 \quad \Rightarrow \quad t = \sqrt{2h/g} = \sqrt{2 \times 20 / 10} = \sqrt{4} = 2.0 \text{ s}$$

Step 2 — the range. Horizontally the speed stays 12 m/s, so in 2.0 s the ball travels

$$\text{range} = 12 \times 2.0 = 24 \text{ m}$$

Step 3 — why the fall time also sets the horizontal distance. Horizontal and vertical motion are independent: the 12 m/s launch does nothing to how fast the ball falls, so the 2.0 s it takes to drop is the same 2.0 s available for horizontal travel. That is what licenses the relation range = speed x time.

$$\text{horizontal distance} = v_x \times t_{\text{fall}} = 12 \times 2.0$$

Step 4 — check the units and the scale. $\sqrt{2h/g}$ has units $\sqrt{\text{m} / (\text{m/s}^2)} = \sqrt{\text{s}^2} = \text{s}$, so the time is dimensionally sound; and a 20 m drop yielding a 2 s fall, stretched by 12 m/s, lands at 24 m -- the right order.

So **Answer: C.**

Step 5 — check the answer. The range should exceed the 20 m height (it does) and be of order the speed times a couple of seconds. 24 m fits.

The distractors.

- **12 m** uses the 12 m/s launch speed as if it were the time of fall.
- **48 m** doubles the correct range, perhaps using $2 t$ or adding the height.
- **34 m** is a muddled blend that sits between 24 and 48.
- **17 m** takes $\sqrt{2h/g}$ as the range and never multiplies by the horizontal speed.

Relevant topics: horizontal projectile; independent components; range = $v_x \times t$.

The trap. Using the 12 m/s launch speed as the time of fall, or doubling the result. The time of fall is $\sqrt{2h/g} = 2.0 \text{ s}$, and the range is horizontal speed (12 m/s) times that time.

Q15 · Two perpendicular forces: the equilibrating force — **Answer E**

What is being tested. Combining two perpendicular forces by Pythagoras, then realising the equilibrium force is equal and opposite to their resultant.

Step 1 — the resultant of the two given forces. They are at right angles, so the resultant R has

$$R^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$R = 5 \text{ N}$$

Step 2 — the equilibrium force. For the body to be in equilibrium the vector sum of all forces must be zero, so the third force must cancel R: same magnitude, opposite direction.

required force = 5 N, opposite to the resultant

Step 3 — resolve the two forces onto axes. Put the 3 N force along x and the 4 N force along y. Their resultant then has components $R_x = 3 \text{ N}$ and $R_y = 4 \text{ N}$ -- a vector sum, not an arithmetic one.

Step 4 — recover the magnitude from the components. Because the components are perpendicular, Pythagoras gives the magnitude:

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{3^2 + 4^2} = 5 \text{ N}$$

So **Answer: E.**

Step 5 — check the size. A vector sum of 3 N and 4 N must exceed each but fall short of their sum 7 N. 5 N does exactly that.

The distractors.

- **7 N** adds the forces as if parallel ($3 + 4$), ignoring the right angle.
- **1 N** subtracts them ($4 - 3$), as if they opposed along one line.
- **12 N** multiplies them, which is not even a force dimensionally.
- **3.5 N** averages them, another way to ignore the geometry.

Relevant topics: resultant of perpendicular forces; Pythagoras; equilibrium as zero vector sum.

The trap. Adding the two forces as parallel (7 N) or subtracting them (1 N) instead of combining them as vectors at right angles. The resultant is $\sqrt{3^2 + 4^2} = 5 \text{ N}$, and the equilibrant is 5 N opposite.

Q16 · A rotating sphere: how latitude changes the centripetal acceleration — **Answer E**

What is being tested. That centripetal acceleration depends on the distance from the rotation axis, which for a point at latitude λ is $R \cos(\lambda)$, not R.

Step 1 — the radius of the circular path. The sphere spins about its vertical axis. A point at latitude λ is on a circle whose radius is its distance from that axis:

$$r_{\text{circle}} = R \cos \lambda$$

(At $\lambda = 0$, the equator, this is R; at the poles it is 0.)

Step 2 — the centripetal acceleration.

$$a = \omega^2 r_{\text{circle}} = \omega^2 R \cos \lambda$$

On the equator the acceleration is $\omega^2 R$. The ratio is therefore

$$\text{ratio} = (\omega^2 R \cos \lambda) / (\omega^2 R) = \cos \lambda$$

Step 3 — derive the latitude radius from the geometry. At latitude λ the point sits at height $R \sin \lambda$ above the equatorial plane, so by Pythagoras its horizontal distance from the axis is $\sqrt{R^2 - (R \sin \lambda)^2} = R \cos \lambda$. That is why the equator ($\lambda = 0$) gives R and the poles give 0 -- the geometry, not a guess.

$$r_{\text{circle}} = \sqrt{R^2 - R^2 \sin^2 \lambda} = R \cos \lambda$$

Step 4 — apply the centripetal formula to that radius. Since $a = \omega^2 r_{\text{circle}}$, substituting the latitude radius gives $a = \omega^2 R \cos \lambda$; dividing by the equatorial value $\omega^2 R$ leaves $\cos \lambda$, with no extra factor.

So **Answer: E**.

Step 5 — check the endpoints. At the equator ($\lambda = 0$) the factor is $\cos 0 = 1$; at the poles ($\lambda = 90$) it is $\cos 90 = 0$. Cosine is the only option that does both, and at 45 degrees it gives $\sqrt{2}/2$ (about 0.71).

The distractors.

- **$\sin \lambda$** uses the wrong trigonometric factor for the distance from the axis.
- **$\cos^2 \lambda$** squares the correct factor, double-counting the latitude dependence.
- **$\sin^2 \lambda$** is the wrong factor squared.
- **1** claims the acceleration is the same everywhere, ignoring latitude entirely.

Relevant topics: circular motion on a rotating sphere; the radius of the latitude circle; centripetal acceleration and latitude.

The trap. Using R instead of $R \cos(\lambda)$ for the radius of the latitude circle, or $\sin(\lambda)$, or squaring. The distance from the spin axis is $R \cos(\lambda)$, so the acceleration scales as $\cos(\lambda)$: 1 at the equator, 0 at the poles.

Q17 · A square lamina with a corner removed: the shift of the centre of mass — **Answer B**

What is being tested. The negative-mass method for a centre of mass: the cut-out is subtracted from the whole, and the remaining lamina is the difference, weighted by area (which stands in for mass for a uniform sheet).

Step 1 — set up coordinates. Let the original square run from $(0, 0)$ to $(2a, 2a)$, so its centre is at (a, a) and its area is $4a^2$. The removed upper-left corner runs from $(0, a)$ to $(a, 2a)$, area a^2 , centre at $(a/2, 3a/2)$.

Step 2 — the remaining lamina. Its area is $4a^2 - a^2 = 3a^2$. By the negative-mass idea:

$$x_{\text{new}} = (4a^2 \times a - a^2 \times a/2) / (3a^2)$$

Step 3 — the horizontal position. Cancel a^2 :

$$x_{\text{new}} = (4a - a/2) / 3 = (7a/2) / 3 = 7a/6$$

Step 4 — the displacement. The original centre was at $x = a$, so the shift is

$$\text{shift} = 7a/6 - a = a/6$$

to the right, toward the remaining material. So **Answer: B.**

Step 5 — check. The cut is a quarter of the area at the upper left, so the centre of mass should move a little to the right and down -- $a/6$ in x is small and in the right direction. (The vertical shift is $a/6$ down as well, by symmetry of the cut.)

The distractors.

- $a/3$ treats the removed corner as a third of the mass rather than a quarter.
- $a/4$ confuses the side of the cut-out with the displacement.
- $a/2$ is the half-side of the whole lamina, not a displacement at all.
- $a/12$ double-reduces the moment and lands a quarter of the $a/3$ error.

Relevant topics: centre of mass; the negative-mass (cut-out) method; uniform lamina as area.

The trap. Treating the removed corner as a third of the mass, or confusing the cut-out's side with the displacement. The corner is a quarter of the area, and the new centre of mass is (original moment minus corner moment) over the remaining three-quarter mass, giving a shift of $a/6$.

Q18 · Two coherent sources: the fringe spacing — **Answer D**

What is being tested. The two-source fringe spacing $\beta = \lambda D / d$, and the discipline of converting every length to the same unit first.

Step 1 — the formula.

$$\beta = \lambda D / d$$

Step 2 — consistent units (metres).

$$\begin{aligned}\lambda &= 500 \text{ nm} = 5.0 \times 10^{-7} \text{ m} \\ d &= 1.0 \text{ mm} = 1.0 \times 10^{-3} \text{ m} \\ D &= 1.5 \text{ m}\end{aligned}$$

Step 3 — substitute.

$$\begin{aligned}\beta &= (5.0 \times 10^{-7} \times 1.5) / (1.0 \times 10^{-3}) \\ &= 7.5 \times 10^{-4} \text{ m} \\ &= 0.75 \text{ mm}\end{aligned}$$

So **Answer: D.**

Step 4 — check. For visible light and a millimetre-scale slit spacing the fringes are sub-millimetre, so 0.75 mm is the right order; a value of several millimetres would mean the formula or units were wrong.

The distractors.

- **0.25 mm** inverts the formula to $\lambda d / D$.
- **1.5 mm** offers the screen distance D itself.
- **2.5 mm** is a unit-muddled value from mixing mm and nm.
- **0.50 mm** swaps λ and d , putting the 0.30 mm separation in the numerator with no D .

Relevant topics: two-source interference; $\beta = \lambda D / d$; unit consistency.

The trap. Inverting the fringe-spacing formula ($\lambda d / D$) or forgetting to convert nm and mm to metres. $\beta = \lambda D / d$, and with $\lambda = 5.0 \times 10^{-7}$ m, $d = 1.0 \times 10^{-3}$ m, $D = 1.5$ m it is 0.75 mm.

Q19 · The critical angle: its sine for glass to air — **Answer A**

What is being tested. The critical-angle condition and, specifically, that its sine is the reciprocal of the refractive index (n_2/n_1), not the index itself.

Step 1 — the condition. At the critical angle the refracted ray grazes the boundary, so the refracted angle is 90 degrees and Snell's law reads

$$n_1 \sin \theta_c = n_2 \sin 90 = n_2$$

Step 2 — glass to air. Here $n_1 = 1.50$ (glass) and $n_2 = 1.00$ (air):

$$\sin \theta_c = n_2 / n_1 = 1.00 / 1.50 = 2/3$$

Step 3 — the value. $2/3$ is about 0.67. So **Answer: A.**

Step 4 — check. The sine must be below 1 (a critical angle exists only because glass is denser than air) and above 0.5 (the critical angle is below 45 degrees since $1/1.5 > 1/\sqrt{2}$). 0.67 sits correctly between.

The distractors.

- **1.50** is the refractive index read directly as the sine, instead of its reciprocal $2/3$.
- **0.50** would be the sine if the glass index were 2.00 rather than 1.50.
- **0.33** comes from a difference error, $(n_1 - n_2)/n_1 = 1/3$, instead of the ratio n_2/n_1 .
- **1.00** says $\sin(\theta_c) = 1$, which would mean there is no critical angle -- but glass to air does have one.

Relevant topics: total internal reflection; the critical angle; $\sin(\theta_c) = n_2/n_1$.

The trap. Giving the refractive index 1.50 directly, or 0.50 (as if the index were 2.0), or $1/3$ from a difference error, or 1.00. The sine is $n_2/n_1 = 2/3$, the reciprocal of the index.

Q20 · A diode and a load resistor: the operating current from the load line — **Answer C**

What is being tested. The load-line construction: a resistor and battery impose a straight line on the diode's I-V plane, and the operating current is where that line meets the device curve -- not a point read off either alone.

Step 1 — the load line. The supply is 1.0 V and the resistor is 100 ohm. In mA, a voltage V across the diode leaves $1.0 - V$ across the resistor, carrying current $(1.0 - V)/100$ amps = $10(1.0 - V)$ mA. So

$$I_{\text{mA}} = 10 (1.0 - V) \Rightarrow V = 1.0 - 0.1 I_{\text{mA}}$$

This line runs from $(V=1.0, I=0)$ to $(V=0, I=10 \text{ mA})$.

Step 2 — the diode segment. Between 0.55 V and 0.82 V the characteristic is straight and passes through 22 mA at 0.8 V. Its slope is

$$I_{\text{mA}} = 22 (V - 0.55) / (0.82 - 0.55) = 22 (V - 0.55) / 0.27$$

Step 3 — intersect. Substitute $V = 1.0 - 0.1 I_{\text{mA}}$ into the diode line:

$$\begin{aligned} I_{\text{mA}} &= 22 (1.0 - 0.1 I_{\text{mA}} - 0.55) / 0.27 \\ &= 22 (0.45 - 0.1 I_{\text{mA}}) / 0.27 \end{aligned}$$

Multiply out: $0.27 I_{\text{mA}} = 9.9 - 2.2 I_{\text{mA}}$, so $(0.27 + 2.2) I_{\text{mA}} = 9.9$, and

$$I_{\text{mA}} = 9.9 / (0.27 + 2.2) \approx 4.0 \text{ mA}$$

So **Answer: C.**

Step 4 — check. At 4.0 mA the resistor drops $0.1 \times 4.0 = 0.4 \text{ V}$, so the diode has about 0.6 V across it -- on the steep part of the curve, consistent with the drawing. The current is well below the 10 mA intercept, as it must be.

The distractors.

- **10 mA** is the short-circuit intercept of the load line, where the diode voltage is forced to zero.
- **22 mA** is the point on the diode curve at 0.8 V, read with no regard for the resistor.
- **2.0 mA** is the current the load line would allow at 0.8 V across the diode ($(1.0-0.8)/100 = 2 \text{ mA}$), not the intersection.
- **6.0 mA** is a value between the two intercepts that is not the true meeting point.

Relevant topics: load-line analysis; the diode I-V characteristic; solving the operating point.

The trap. Reading the 22 mA point off the diode curve, or the 10 mA short-circuit intercept, as the answer. The current is set by the intersection of the load line $V = 1.0 - 0.1 I_{\text{mA}}$ with the diode segment, which is about 4 mA.

Q21 · An RC charging circuit: the time constant — **Answer C**

What is being tested. The RC time constant $\tau = RC$, and the cancellation of the mega- and micro- prefixes.

Step 1 — the formula. For a resistor and capacitor in series the time constant is

$$\tau = R C$$

Step 2 — base units.

$$R = 1.0 \text{ megohm} = 1.0 \times 10^6 \text{ ohm}$$

$$C = 2.0 \text{ microfarad} = 2.0 \times 10^{-6} \text{ F}$$

Step 3 — multiply. The 10^6 and 10^{-6} cancel:

$$\tau = (1.0 \times 10^6) \times (2.0 \times 10^{-6}) = 2.0 \text{ s}$$

So **Answer: C.**

Step 4 — check. A megohm times a microfarad is a second exactly, so $1.0 \times 2.0 = 2.0 \text{ s}$ is the right order; the supply voltage does not enter the time constant at all.

The distractors.

- **12 s** multiplies the capacitance by the voltage ($2.0 \times 10^{-6} \times 6.0$ with stray factors).
- **3.0 s** uses $R/C = 1.0 \times 10^6 / 2.0 \times 10^{-6}$ inverted into an enormous number mishandled.
- **0.50 s** divides the two, cancelling the prefixes the wrong way.
- **1.0 s** returns the resistance in megohms as the time.

Relevant topics: the RC time constant; $\tau = R C$; prefix cancellation.

The trap. Multiplying the capacitance by the supply voltage, or using R/C instead of R times C . The time constant is $R C$, and 1.0 megohm times 2.0 microfarad is exactly 2.0 s ; the voltage is irrelevant.

Q22 · Melting ice: the time from energy = power x time and latent heat — **Answer E**

What is being tested. The melting-energy relation $E = m L$ and the heater relation $E = P t$, combined into $t = m L / P$.

Step 1 — the energy to melt.

$$E = m L = 0.20 \times 3.3 \times 10^5 = 6.6 \times 10^4 \text{ J}$$

Step 2 — the heater rate. A 1.0 kW heater supplies $P = 1000 \text{ J/s}$.

Step 3 — the time.

$$t = E / P = 6.6 \times 10^4 / 1000 = 66 \text{ s}$$

So **Answer: E.**

Step 4 — check. A kilowatt for about a minute melting a few hundred grams of ice matches everyday experience (a kettle melts far less), so the scale is right.

The distractors.

- **33 s** forgets to divide by the power, offering the energy 6.6×10^4 trimmed by a factor of two.

- **132 s** doubles the mass or the latent heat.
- **200 s** mishandles the 1000 W and inflates the time threefold.
- **6.6 s** divides by 1000 twice.

Relevant topics: latent heat of fusion; $E = m L$; heater power and time.

The trap. Forgetting to divide by the heater power (so the energy in joules is offered as seconds), or doubling the mass/latent heat. The time is $m L / P = 66 \text{ s}$ with $P = 1000 \text{ W}$.

Q23 · Mass-energy: energy released from a mass defect — **Answer B**

What is being tested. The mass-energy equivalence as a single multiplication, using the convenience that one atomic mass unit converts to 930 MeV.

Step 1 — the principle. Whenever mass is lost, the released energy is

$$E = m c^2$$

Step 2 — the shortcut. Rather than carry c^2 , the problem gives $1 \text{ u } c^2 = 930 \text{ MeV}$. So a defect of 0.20 u gives

$$E = 0.20 \times 930 = 186 \text{ MeV}$$

Step 3 — carry the units through the multiplication. The 930 MeV already bundles c^2 into per u, so the only arithmetic is the mass in u times that rate; keeping the u in view stops the defect (0.20 u) being confused with a whole atom.

$$\text{rate} = 930 \text{ MeV} / \text{u} ; E = 0.20 \text{ u} \times 930 \text{ MeV/u} = 186 \text{ MeV}$$

Step 4 — confirm the figure is consistent. The 930 MeV per u is itself a rounded figure (the exact $1 \text{ u } c^2$ is 931.5 MeV), so 186 MeV is the right magnitude; quoting more digits would overstate the precision of the given constant.

So **Answer: B**.

Step 5 — check. The answer must be less than 930 MeV (one whole u) but far more than a few MeV, and 186 MeV sits correctly between -- a typical scale for a single nuclear event.

The distractors.

- **93 MeV** uses only 1 u of mass, ignoring the 0.20 factor.
- **372 MeV** doubles the defect to 0.40 u.
- **46 MeV** divides by the equivalent ($930 / 0.20$) the wrong way.
- **930 MeV** is the factor itself, the energy of one u, with the mass forgotten.

Relevant topics: mass-energy equivalence; $E = m c^2$; the $1 \text{ u} = 930 \text{ MeV}$ shortcut.

The trap. Using 1 u of mass (93 MeV) or doubling to 0.40 u (372 MeV), or returning the 930 MeV conversion factor itself. The released energy is the defect times the equivalent: $0.20 \times 930 = 186 \text{ MeV}$.

Q24 · The photoelectric effect: maximum kinetic energy of the electrons — **Answer D**

What is being tested. The photoelectric equation $K_{\text{max}} = hf - \phi$: the photon's energy pays the work function first, and whatever is left is the electron's kinetic energy.

Step 1 — the equation.

$$K_{\text{max}} = hf - \phi$$

Step 2 — substitute the given energies. The photon energy is $hf = 5.0 \text{ eV}$ and the work function is $\phi = 2.0 \text{ eV}$:

$$K_{\text{max}} = 5.0 - 2.0 = 3.0 \text{ eV}$$

Step 3 — the energy book-keeping. The photon arrives with 5.0 eV and the metal demands 2.0 eV to free the electron, so the electron leaves with the remainder -- a strict subtraction, not a shared or added amount:

$$K_{\text{max}} = E_{\text{photon}} - \phi = 5.0 - 2.0$$

Step 4 — the emission condition. Emission requires the photon energy to exceed the work function; here $5.0 \text{ eV} > 2.0 \text{ eV}$, so the leftover 3.0 eV is genuinely available as kinetic energy, and any answer at or above 5.0 eV would break the energy budget.

So **Answer: D**.

Step 5 — check. The kinetic energy must be positive (emission happens) and below the photon energy (the work function was paid). 3.0 eV satisfies both.

The distractors.

- **2.0 eV** is the work function -- the energy that had to be spent, not what is left.
- **5.0 eV** is the photon energy, with no subtraction at all.
- **7.0 eV** adds the two, as if they reinforced.
- **1.0 eV** subtracts the wrong way, $\phi - hf$, giving a negative (which would mean no emission).

Relevant topics: the photoelectric effect; $K_{\text{max}} = hf - \phi$; work function.

The trap. Offering the photon energy (5.0 eV) or the work function (2.0 eV), or adding them (7.0 eV). The kinetic energy is the photon energy minus the work function: $K_{\text{max}} = 5.0 - 2.0 = 3.0 \text{ eV}$.

Q25 · A hydraulic jack: the load lifted by the larger piston — **Answer A**

What is being tested. Pascal's principle in a hydraulic jack: equal pressure in both pistons means the force is multiplied by the area ratio.

Step 1 — equal pressure.

$$F_{\text{small}} / A_{\text{small}} = F_{\text{large}} / A_{\text{large}}$$

Step 2 — solve for the large force. Rearranging the pressure balance gives the force on the large piston as the small force multiplied by the area ratio:

$$F_{\text{large}} = F_{\text{small}} \times (A_{\text{large}} / A_{\text{small}})$$

Step 3 — the area ratio. Because it is a ratio, the cm^2 units cancel:

$$A_{\text{large}} / A_{\text{small}} = 40 / 2.0 = 20$$

Step 4 — the load.

$$F_{\text{large}} = 100 \times 20 = 2000 \text{ N}$$

So **Answer: A.**

Step 5 — check. The jack multiplies the input force by 20, which is its purpose; the result is far larger than 100 N, as it must be, and no conversion was needed since both areas are in cm^2 .

The distractors.

- **100 N** is the applied force, ignoring the area multiplication entirely.
- **400 N** squares the gain -- using 20 and then multiplying by 20 again.
- **50 N** divides the input force by the small area, giving a pressure, not a force.
- **500 N** inverts the area ratio to $1/20$ and shrinks the force.

Relevant topics: Pascal's principle; hydraulic machines; force scales with area.

The trap. Giving the applied 100 N as the answer, or squaring the area ratio ($20 \times 20 = 400 \text{ N}$), or inverting it. The force multiplies by the area ratio: $F_{\text{large}} = 100 \times (40/2.0) = 2000 \text{ N}$, and the cm^2 units cancel in the ratio.